

Raw phi1 voltage calibration

RUN

194748_raw-phi1_connection-check

Status: completed; 3 individual PAX readings

MEASUREMENT

0.0 to 0.8 RP command V, step 0.02 V; 3 forward/reverse cycles.

Settling: 1.0 s; 3 consecutive readings per held level.

VOLTAGE GRID

Uniform grid specified above.

PHYSICAL SETUP

Physical position awaiting confirmation; no assumed port

IN1/IN2 physical mapping awaiting confirmation

WHAT IS RECORDED

Requested RP voltage, ASG register readback, full FPGA OUT1 scope trace, explicitly selected IN1/IN2 scope traces, raw PAX records and acquisition times.

WHAT IS NOT ESTABLISHED

Internal OUT1 data is the signal before the DAC. It is not an independent BNC or piezo-terminal voltage measurement. No historical driver gain is applied.

ANALYSIS POLICY

No curve fits, interpolated V_{π} , drift subtraction, or optimized coordinate rotation. Phase is $\text{atan2}(S3, -S2)$; unwrapping only adds 2π at coordinate wraps. Step medians are labelled summaries; all individual measurements remain in data.csv and pax_raw.jsonl.

DATA VALIDITY NOTE

digital_out1_v/digital_out2_v and cleanup_digital_out*_v are INVALID: Python register definitions at 0x164/0x168 conflict with FPGA trigger-timestamp registers. Do not interpret those fields as voltages.

Raw phi1 voltage calibration

DATA VALIDITY NOTE

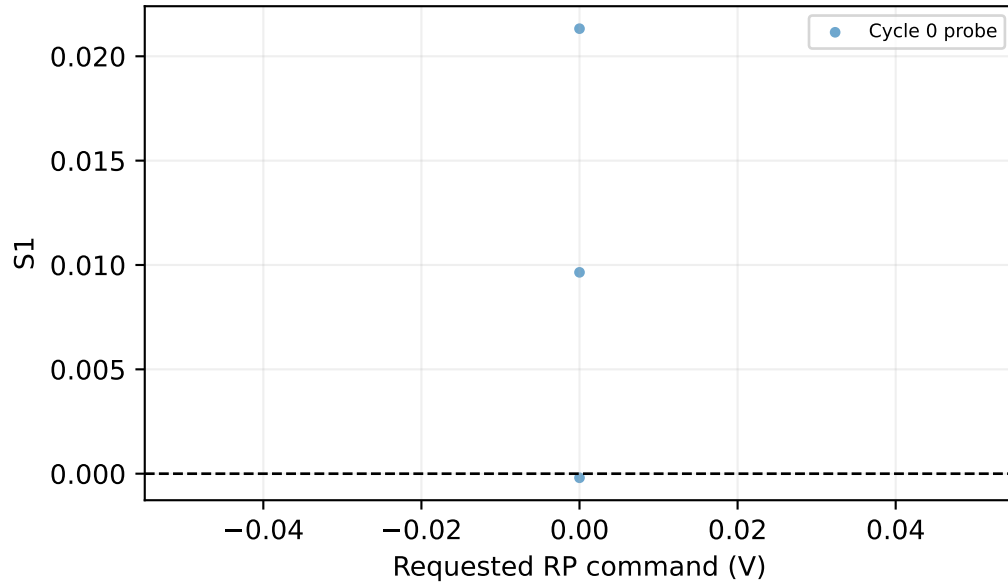
The saved scope/0000.npz out1 channel is valid and reads zero throughout. Subsequent runs use recorded scope traces rather than the conflicting registers.

DATA VALIDITY NOTE

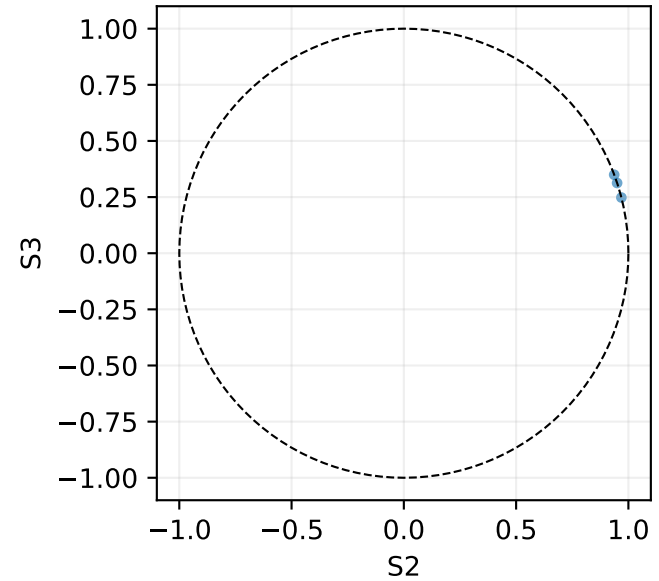
CSV analog_in1_v also follows the selected scope source out1; it is not a physical IN1 measurement. No verified terminal-voltage measurement was acquired.

Individual PAX readings at held voltages

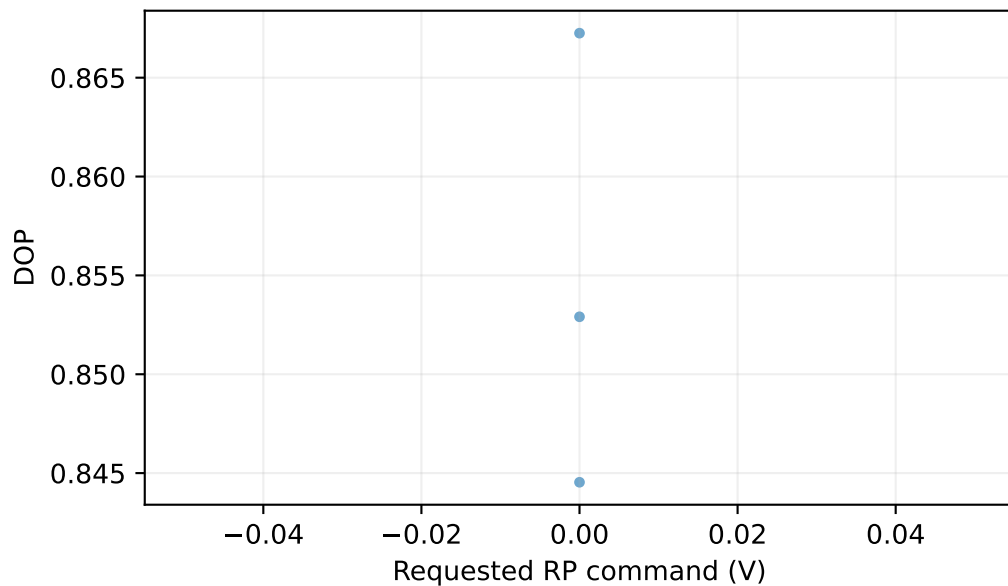
Departure from the $S_1 = 0$ plane



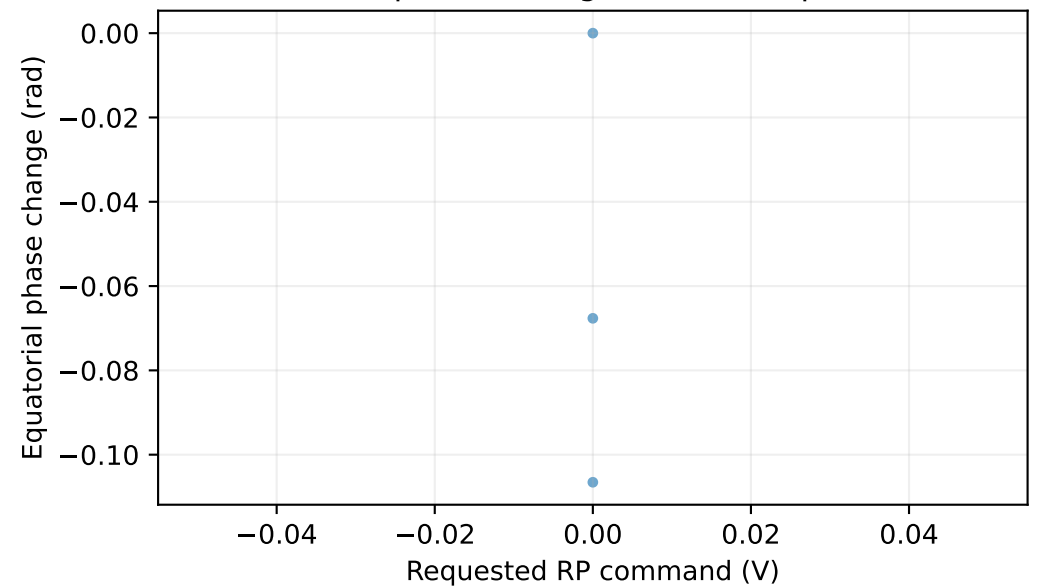
Raw orientation trajectory



Raw PAX DOP



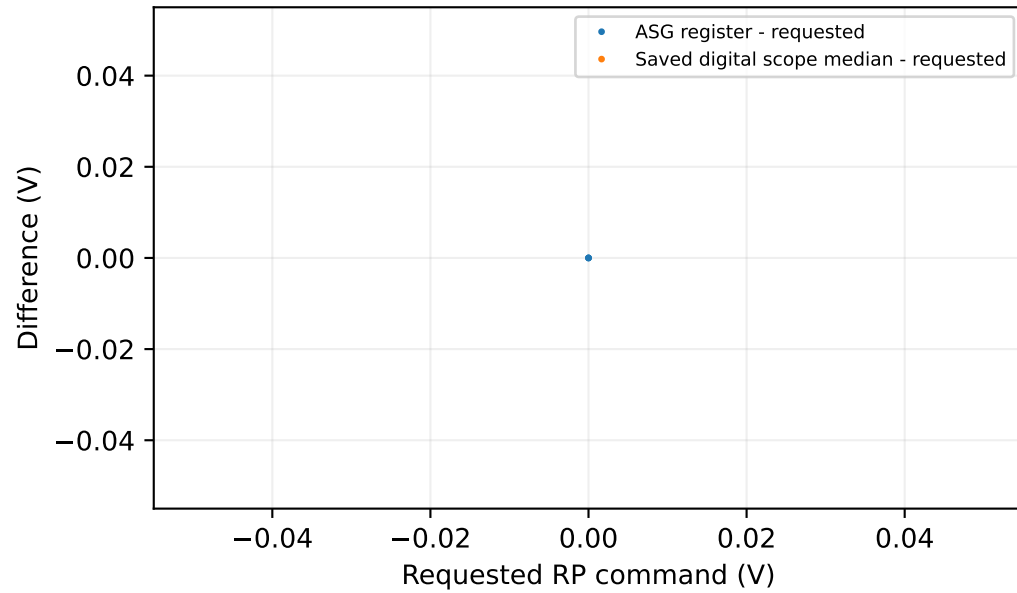
Measured phase change from each pass start



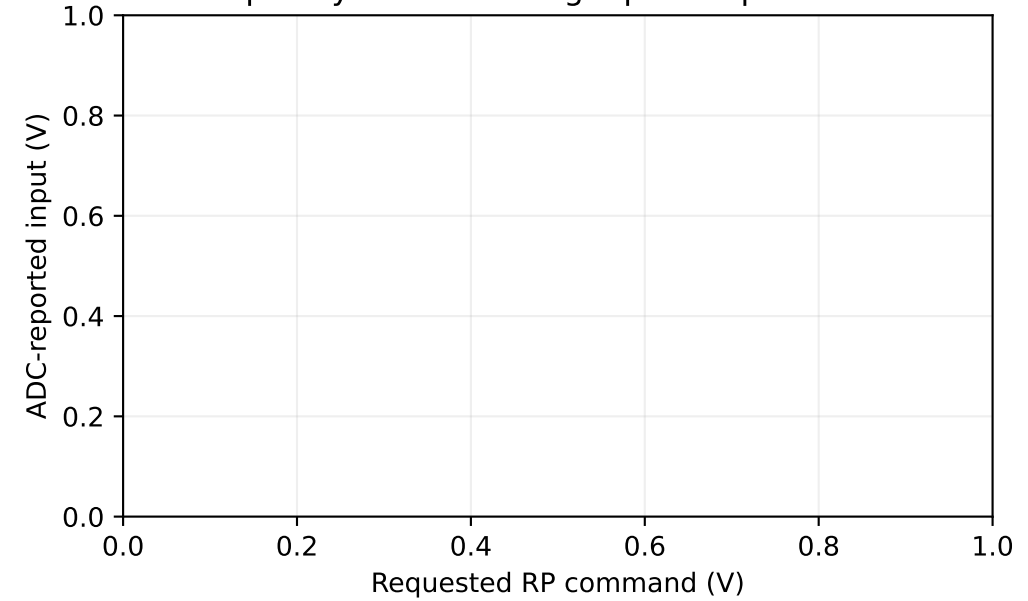
Orientation Stokes are unit-normalized from PAX angles; DOP is shown separately. No fitted or corrected trajectories.

Readbacks and direct return-path comparisons

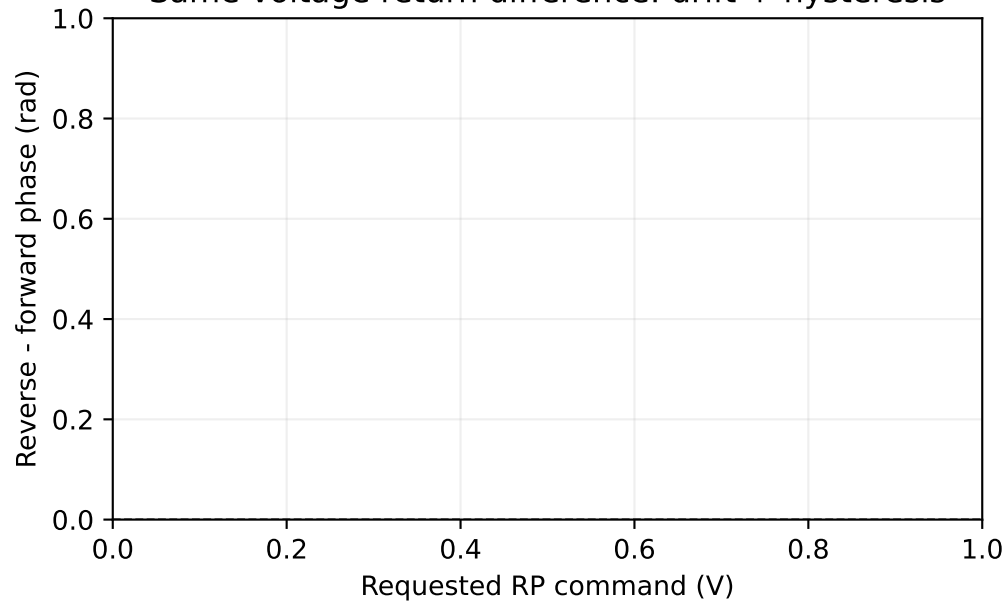
Software command versus internal readbacks



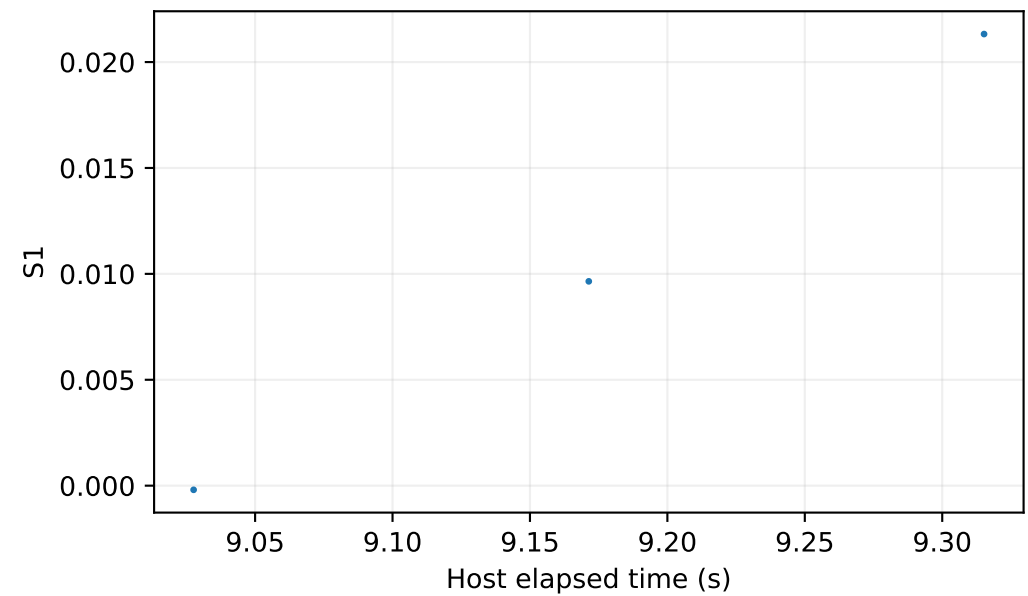
Explicitly routed analog input scope records



Same-voltage return difference: drift + hysteresis



Time record — no drift subtraction



Observed pi-crossing brackets

INTERPRETATION

Each interval brackets π between adjacent measured step medians; no interpolation is applied. Drift, hysteresis and finite step size remain in the result. These are RP command-voltage brackets for projected equatorial phase, not calibrated terminal volts or an independent measurement of actuator retardance. Departure from $S1=0$ limits the ideal-model interpretation.

Active acquisition parameters

PROBE

True

START

0.0

STOP

0.8

STEP

0.02

REPEATS

3

SETTLE

1.0

SAMPLES

3

LABEL

connection-check

PAX_LOCATION

Physical position awaiting confirmation; no assumed port

INPUT_CONNECTIONS

IN1/IN2 physical mapping awaiting confirmation

PAX SETTINGS

Wavelength 830.0 nm; daemon mode 5; configured rotation 60.0 Hz

Active acquisition parameters

CLEANUP EVIDENCE

not available